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Computer science Department



Software Engineering
Term 1 – 2025/2026

Software Project Management Plans

For

Sanad platform

Version 1.0

CIS Year 4, G 4

Dr.Enas El-Sharawy

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This Software Project Management Plans (SPMP) was prepared and provided as a deliverable for [Course Name, number, term], and it will be used by [name of end user].

This document is based in part on the IEEE Recommended Practice for SPMP Descriptions.

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Revision History

Name	Date	Reason For Changes	Version
All members	Oct. 9, 2025	Prepared initial version	0.1
All members	Oct. 10, 2025	Updated section 1	0.2
All members	Oct. 11, 2025	Updated section 2	0.3
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1. Project Overview

1.1 Purpose, Scope, and Objectives

Sanad platform project targets the establishment of an online interactive platform designed for people with disabilities with the objective of facilitating access to the required services and information and social and professional integration.

The site brings together several functions in one friendly and understandable interface, which includes interactive maps showing available sites, a community forum allowing the possibility for communication, and helpful electronic assistance, such as volunteering and interpreter requests, in addition to learning and job prospects.

Scope:

The project includes the following constituents:

- 1.Designing an interactive map indicating publicly available sites for disabled people such as universities, hospitals, and restaurants.
- 2.Create a community section to exchange experiences and inspiring stories among users.
- 3.Provide online services for ordering and requesting volunteers or sign-language interpreters.
- 4.Develop an education and career section that connects the users with the appropriate courses and job vacancies.
5. Create a database of assistive technologies, for example screen reader software, speech command devices and smart canes.

Out of Scope:

The initiative does not encompass the creation of tangible assistive technologies or hardware elements. Furthermore, it excludes monetary exchanges, as the system is designed solely to disseminate information and link users to currently available services.

Objectives:

- Enhance the accessibility of the facilities and services for disabled people.
- Promote social interaction and community integration.
- Provide volunteer and relief services for daily needs.
- Support access to school and job training for individuals with disabilities.

- Guarantee usability, multi-language access, and compliance with global accessibility guidelines (WCAG 2.1).

1.2 Assumptions, Constraints and Risks

Assumptions:

- Users have access to the internet and gadgets (PCs/smartphones) in order to use the platform.

The initiative will be executed utilizing open-source software.

- The operating environment shall be in an academic environment, with possible cloud-based simulation (e.g., AWS) for testing.

Constraints:

- The project's lifespan extends only for a single semester.
- The size and technical capabilities are limited.
- Each deliverable should follow the course instructor's announced deadlines.

Risks:

- Disruptions due to coordination failures among team members.
- Problems with integrating the different system parts (mapping, discussion board, services).
- Difficulty in implementing features for multiple types of disabilities.
- Data security and user-privacy risks during testing and deployment.

1.3 Project Deliverables

Deliverable	Deadline (Week)
Defining Your Project for the Course (Project Suggestions)	By the end of week 3
Submit Project Proposal	By the end of week 4
Submit Project Management Plan (SPMP)	By the end of week 7
Submit a status report 1	By the end of week 8
Submit Project Requirements (SRS)	By the end of week 10
Submit Project Design (SDS)	By the end of week 11
Submit a status report 2	By the end of week 12
Submit Project Test Plan (STS)	By the end of week 13
Delivery of the prototype and presentation of the project	By the end of week 15

Table 1.1- Project Deliverables

1.4 Schedule and Budget Summary

Schedule Summary:

Weeks 3–4: project proposal; Weeks 5–7: SPMP;

Weeks 8–13: requirements, design, and testing;

Weeks 14–15: prototype and presentation.

Budget Summary:

Financial Statement: This undertaking is entirely academic in nature and incurs no financial expense. All software and tools used, such as the development environments, frameworks, versioning tools, and test servers, are open-source software or educational license software.

1.5 Evolution of the Plan

This SPMP document is a dynamic document and shall be updated whenever significant changes are incorporated in the project scope, activities, or team configuration. Each update shall remain version-tracked and stored in the shared team repository. The continual

redefinition of this document ensures its compliance with the IEEE 1058 standard and maintains effective project management throughout the software lifecycle.

1.6 References

[1] IEEE Computer Society, IEEE Standard for Software Project Management Plans (IEEE Std 1058-1998), New York, NY, USA: IEEE, 1998.

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[12] A. Madugalla, Y. Huang, J. Grundy, M. H. Cho, L. Gamage, T. Leão, and S. Thiele, “Engineering Adaptive Information Graphics for Disabled Communities: A Case Study with Public Space Indoor Maps,” 2024.

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1.7 Definitions and Acronyms

- SPMP: Software Project Management Plan
- SRS: Software Requirements Specification
- SDS: Software Design Specification
- STS: Software Test Specification
- UI/UX: User Interface / User Experience
- WCAG: Web Content Accessibility Guidelines
- UAT: User Acceptance Testing
- AWS: Amazon Web Services (used for prototype cloud deployment simulation).

1.8 Document Structure

1. Project Overview.
2. Project Organization.
3. Managerial Process Plans.
4. Technical Process Plans.
5. Supporting Process Plans.
6. Additional Plans.

2. Project Organization

2.1 External Interfaces

In this part we describe both the organizational boundaries and the external relationships involving the Sanad project. Many entities are collaborating toward successful planning, implementation and delivery of the platform vision.

The work is coordinated through the supervision of Dr. Enas El-Sharawy who is the Project Advisor and holds a role in the parent organisation for oversight in an academic and managerial role. The Project Team is composed of student members: Khawla Almoalm, Fatimah Alawa, Zainab Esaeed, Zahra Alsadah, Jood Alzayer, Fatimah Almur, and Remas Alqahtani. An internal Project Team manages the software design, development, documentation and testing phases throughout the project.

The project is externally engaged with two categories of organizations:

1. External Service Providers:

These organizations offer the necessary technology and infrastructure support needed for the Sanad platform. These are :

- Map & Location Services to support accessibility mapping and geographic data.
- Payment Gateways for secure transactions online.
- Video Communication Services to provide remote communication and consultations.
- Cloud & Hosting Services for scalability and reliability.
- Assistive Technology Providers to create tools and devices to enhance user access .
- Governmental Entities to work together, with compliance and data validation.

2. End Users / Customers:

The following individuals are the key audience and beneficiaries of the Sanad platform :

- People with Disabilities (main users) who are directly interacting with the services available in the platform.
- Families and Caregivers who support them with day-to-day needs and accessibility tasks.
- Volunteers and Service Providers who contribute to support and community engagement.

- Organizations and Establishments that advance access awareness.
- Government Entities that regulate and/or support access initiatives.

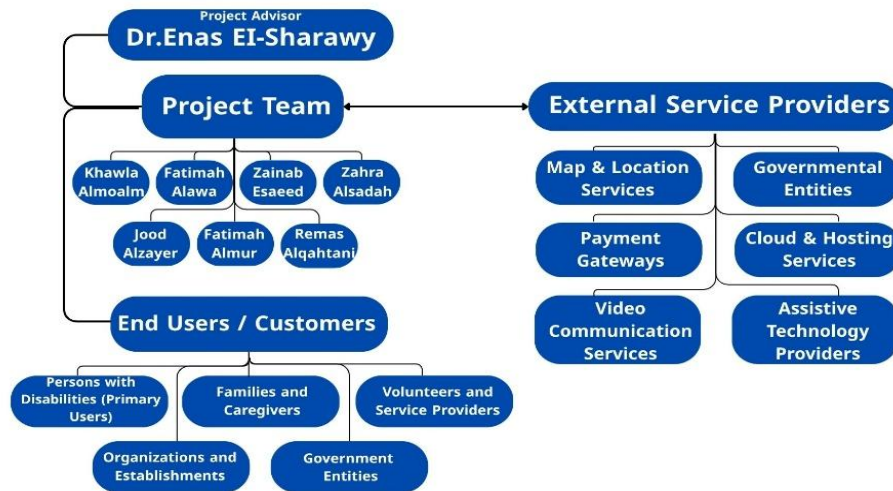


Figure 2.1 – External Interfaces

2.2 Internal Structure

This section outlines the internal organizational structure of the "Sanad" project and outlines the lines of authority, communication and accountability within the project team.

The project is led by Dr. Enas El-Sharawy, who serves as the Project Advisor and provides academic and managerial support to the project. The advisor is responsible for ensuring that the project meets institutional standards, monitoring progress, and verifying main deliverables.

Khawla Almoalem is the Project Leader and will oversee all project phases and support the development team. The project leader will be the point of communication between the advisor and the rest of the team to ensure timelines, quality and goals are all maintained.

The Development Team comprises:

Fatimah Alawa, Zainab Esaeed, Zahra Alsadah, Jood Alzayer, Fatimah Almur, and Remas Alqahtani.

Each team member has responsibility for a distinct set of development, testing, documentation, or design tasks. Team members communicate with one another based on regularly scheduled meetings, shared documentation spaces, and collaborative development tools.

The internal structure of the project also supports formal coordination with institutional entities which support processes including quality assurance, verification, validation, and configuration management in partnership with the project advisor.



Figure 2.2 – Internal Structure

2.3 Roles and Responsibilities

This section outlines the major work activities, supporting processes on the Sanad project, and the organization for each role in each unit. Where needed, project roles are assigned specific responsibilities to maintain communications, accountability, and quality through all phases of development.

The Project Leader shall maintain coordination and communication among team members and have direct communication with the academic supervisor. All members of the team work together during project planning, requirements analysis, documentation, and presentations, while the individual's specialized roles, i.e., Designer, are focused on the design of the user interface and user experience (UI/UX).

Role	Responsible members	Responsibilities
Project planning	All members	Define project objectives, scope of work, timeline, and task allocation.
Requirements analysis	All members	Collect and analyze the needs of users with disabilities.
Interface design	Designer	User Interface (UI/UX) Planning.
Documents	All members	Preparing project documents (SRS, SPMP, SDS) and final reports.
Communication and meetings	Leader	Coordinate meetings, ensure continuous communication with the academic supervisor, and submit periodic reports.
Preparing presentations	All members	Design and prepare presentation slides for the introductory presentation and final project presentation.

Table 2.3 – Roles and Responsibilities

3. Managerial Process Plans

This section describes the management processes that will be implemented as part of the Sanad Project. These procedures include starting, work planning, tracking, risk management, and project close-out techniques. The primary goal of these processes is to provide effective control, communication and a successful completion of all project objectives while adhering to the established scope, schedule and quality restrictions.

3.1 Startup Plan

3.1.1 Estimates

The Sanad project is planned to span 14 weeks, from September to December 2025. Free and open-source tools like GitHub, Trello, Visual Studio Code and Figma are used in this project. Free and open-source tools like GitHub, Trello, Visual Studio Code and Figma are used in this project.

- An assurance level of 85% is anticipated for this project based on data from prior student projects.
- Among the estimate techniques used are:
 - Analogy-based estimation, which makes use of information from related academic studies.
 - Historical approximation based on knowledge gained from earlier accessibility initiatives.
- At every significant stage, including design, implementation and testing, estimates will be examined. Gantt charts for comparative analysis and Trello for progress tracking are examples of re-estimation tools.

3.1.2 Staffing

In staffing activities, we specify the number of team members and their levels of knowledge/skills required in each phase of the project. The Sanad project team includes Khawla Almoalem (Leader), Fatimah Alawa, Zainab Esaced, Zahra Alsadah, Jood Alzayer, Fatimah Almur, and Remas Alqahtani.

Phase	Team Members	Skills / Roles	Duration
Requirements and Design	3	Analyst and UI/UX Designer	2 weeks
Development	3	Front-end and Back-end Developers	6 weeks
Testing and Documentation	2	QA Engineer and Documentation Writer	3 weeks

Table 3.1.2- Staffing

Everyone on the team will be engaged in every phase of the project and the entire team will be involved in the analysis, design, development, testing and writing of documentation as shown on Table [3.1.2]. Staffing is internal (student-based) and is supervised by Dr. Enas El-Sharawy.

3.1.3 Project Staff Training

The training programs are designed to improve both technical and managerial skills among team members.

Training Topic	Participants	Methods	Source / Platform
Accessibility UI/UX Design	3	Online tutorials	Coursera or YouTube
Project Management with Trello	6	Self-learning	Trello Guides
Software Testing and Quality	2	Mentoring	Internal (Project Leader)

Table 3.1.3 - Project Team Training

This training will guarantee that all staff members are proficient in critical areas such as accessibility design, agile teamwork approaches, and quality control procedures.

3.2 Work Plan

3.2.1 Work Breakdown Structure

The Work Breakdown Structure (WBS) separates the Sanad project into discrete work units. Each of those units has defined objectives, resources required, time estimates, deliverables, and acceptance.

Having the work broken down allows for precise estimates of time, cost and risk for each stage as well as the ability to track and call out accountability for all levels of work. The activities are collected into the following general categories:

- Analysis Phase - Gathering and validating requirements for disability and accessibility needs.
- Design Phase - UI/UX design, architectural design, and database design.
- Development Phase - Development of unique modules such as mapping, community, services, and learning.
- Testing Phase - System integration and validation as well as accessibility testing.
- Documentation & Delivery - Creating SRS, SDS, SPMP and final report.

3.2.2 Schedule Allocation

- The schedule outlines the order and relationships among all work activities.
- Dependencies, milestones, and deadlines are identified to show opportunities for parallel development.
- The project schedule indicates the critical path, which is a sequence of tasks that directly affects the overall duration.
- Milestone reviews will also be used to assess progress, quality, and scope.
- The techniques used for visualizing time dependencies and progress tracking include Gantt charts, activity networks, and PERT charts.

3.2.3 Resource Allocation

Resource allocation is a detailed identification of resources aligned with each work activity.

The project will utilize human and technical resources, including:

- People: Members of the team will be assigned according to skill level and task type.
- Computing Resources: Laptops, accessibility testing tools, and development environments.
- Software Tools: Figma (design), Visual Studio Code (development), and Trello (project tracking).
- Administrative Support: Access to institutional facilities, feedback from advisers, and documentation systems.

3.2.4 Budget Allocation

The project budget outlines the estimated costs of all key work activities. Although the Sanad project is utilizing an academic environment and will not incur actual costs, the hours of effort and use of tools are charged as costs equivalent to using a costed resource. The budget estimation includes:

- Computing and software resources.
- Internet and hosting services.
- Meetings and training sessions.
- Administrative and documentation support.

Each resource subject to charges is described as a line item in the WBS to clarify tracking and evaluation.

3.3 Project Tracking Plan

The Project Tracking Plan outlines the procedures and methods for tracking the Sanad project's progress. It explains how the project will plan for measuring, monitoring, and reporting on requirements, schedules, quality, reports, and metrics.

3.3.1 Requirements Management

- All requirements are documented in the SRS and tracked using GitHub Issues.
- Any change request will be reviewed and approved by the project leader and advisor.
- A Requirements Traceability Matrix (RTM) links requirements to corresponding test cases.

3.3.2 Schedule Control

Progress is tracked weekly using Trello dashboards.

Delays are reduced by shifting jobs or extending minor milestones.

Progress indicators include:

- Percentage of tasks completed
- Open/closed task ratio
- Milestone adherence

3.3.4 Quality Control

Quality control is maintained through:

- peer code reviews
- Unit and integration testing
- Adherence to IEEE 829 testing requirements.
- Advisor verification of deliverables prior to submission

3.3.5 Reporting

The project's internal reporting rules control all reporting methods, formats, and communication channels. A weekly report includes accomplishments, schedule compliance, budget status, and highlighted risks. Trello, Google Drive, Microsoft Teams, and email are the communication channels.

3.3.6 Project Metrics

These indicators are intended to assure timely delivery, limit costs, maintain high quality, and constantly improve project performance.

Matric	Description/Importance	Goal
Time	The amount of time required to perform each assignment by the deadline.	To fulfill project deadlines, complete tasks as quickly and as efficiently as feasible.
Cost	The project's budget includes costs for vehicles, tools, hardware, and so on.	Maintain a budget and make sensible resource allocations.
Quality	The quality of deliverables given to clients, senior management, and the project manager.	Maintain high-quality deliveries that meet or exceed expectations.
Performance & improvement	Continuously monitoring growth of teams and project success.	Evaluations are conducted on a regular basis to guarantee continuous development and project success.

Table 3.3.6 – Project Metrics

3.4 Risk Management Plan

As a risk management strategy for the "Sanad platform" project, this table lists the most significant potential risks that our team might come across during the development and implementation stages. In addition to a plan to deal with the risk should it materialize, the table also details the type of risk, its impact on the project, and its likelihood of occurring. The team uses this analysis to anticipate possible risks and lessen their effects on the project in order to guarantee effective continuity and a successful outcome.

Risk	Type	Effects	Probability	Example Plan
Inadequate courses and tools	Technical	Lack of sufficient training materials may reduce efficiency and delay development.	Moderate	Collaborate with other learning platforms and provide alternative resources.
Emergency circumstances	People	Team member unable to complete tasks on time, causing work overload for others.	Low	Establish an emergency plan and redistribute work among team members.
Requirements redistribution	Requirements	Changes in requirements cause confusion and delays in implementation.	Moderate	Prioritize and validate new requirements with the advisor before implementation.
Dropping the course	Organizational	Loss of experienced member halts project progress and causes delays.	Low	Redistribute tasks and prepare an alternative plan to cover responsibilities.
Late submission	Estimation	Delayed deliverables result in reduced academic performance and project evaluation.	High	Monitor progress weekly and perform follow-ups using Trello and meetings.

Table 3.4 Risk Management Plan

3.5 Project Closeout Plan

The table below details the essential actions necessary for the formal closure of the project. Each task is associated with its respective objective to guarantee an organized transition, adequate documentation, and efficient knowledge retention.

Description	Current Point
To ensure team engagement and facilitate a smooth transition.	After the knowledge transfer, reassign project staff and close contractor positions.
To save all project assets for later usage.	All papers, drawings, reports, and source code should be archived in the central storage system.
Documenting performance results.	Keep track of crucial indicators such as schedule, cost, user acceptance, and accessibility success.
Gather feedback and thoughts from the team.	Conduct a post-project debriefing with every team member.
To officially complete the project.	Prepare a final report outlining your accomplishments and lessons learned.

Table 3.5 Project Closeout Plan

4. Technical Process Plans

4.1 Process Model

This section summarizes the process model that was implemented through the Sanad project.

- This project followed an iterative and phased approach, with development and review activities occurring simultaneously to ensure quality and user-focused design.
- Each phase has explicit deliverables that are inputs for the next phase, while review and advisor approval took place in a manner to help maintain project momentum.
- The accepted academic process model was adapted to implement a special emphasis on accessibility assessment and inclusive design.

Phase	Description
Initiation	We began by outlining the project's primary objective and comprehending the requirements of individuals with impairments.
Planning	Plan the platform's primary features and analyze the needs.
Design	Make prototypes and wireframes with Canva and Figma.
Review	Get input and make design improvements.
Finalization	Send in the completed, authorized prototype and supporting materials.

Table 4.1.1 - Process Model

Milestone	Deliverable
M1	Project idea approval
M2	Prototype completed
M3	Final design submission

Table 4.1.2 - Process Model

4.2 Methods, Tools, and Techniques

This chapter describes the development methods, tools, and techniques utilized for the Sanad project.

- The system used a user-centered and iterative design methodology and focused on accessibility, usability, and quality.
- Development activities included requirements gathering, prototype design, team review, testing, and final delivery.
- A variety of tools were adopted for design, documentation, and presentation, including Canva, Figma, Google Docs, and PowerPoint.

Procedure Step	Details
Step 1	Collect user requirements and feedback
Step 2	Create and test prototypes for usability
Step3	Conduct team review and apply improvements
Step 4	Finalize and submit the platform design

Table 4.2.1 - Methods, Tools, and Techniques

Category	Description
Approach	User-Centered and Iterative Design focused on accessibility
Tools & Software	Canva, Figma, Google Docs, PowerPoint
Processes	1-Design 2- Review 3-Test 4-Improve 5- Finalize
Standards	WCAG 2.1 Accessibility Guidelines.
Performance	Simple and organized design reports
Policies	More then 85% positive feedback from pilot users via survey or live trial.
Documentation	Complete and accurate user manuals, technical documentation, and deployment guides are delivered.

Table 4.2.2 - Methods, Tools, and Techniques

4.3 Infrastructure

The operational foundation offers the necessary environment and resources needed to successfully implement the Sanad system. The operational foundation consists of the hardware, software, operating systems, network configurations, and facilities that support analysis, design, development, testing, and deploying of the system.

The foundation establishes an environment that is secure, stable, and operationally efficient for all project team members to encourage collaboration, safeguard data, and support the adherence to project specific standards and policies.

Category	Description
Hardware	Workstations for developers and testers, network routers, and cloud servers (AWS or Azure).
Operating System	Windows 10/11 and Ubuntu Linux for development, testing, and deployment.
Network	Secure LAN for internal communication and VPN for remote access.
Software	Development tools: Visual Studio Code, Android Studio, MySQL Workbench. Testing tools: Selenium, Postman. Project management: Trello, Jira.
Policies and Procedures	Security policies including role-based access control, data encryption, and daily backups. Version control using GitHub.
Standards	Coding, documentation, and testing standards following IEEE and WCAG accessibility guidelines.
Facilities	Fully equipped office space with desks, meeting rooms, administrative support, and janitorial services. Physical security measures in place to protect assets and personnel.

Table 4.3 – Infrastructure

4.4 Product Acceptance

In this subsection, the plan and objective criteria for acceptance of the Sanad project deliverables will be clarified.

- Each deliverable will be assessed and evaluated against pre-defined measureable acceptance standards in order to certify that it meets the specifications for all required aspects of functionality, quality, security, and accessibility.
- Acceptance testing and evaluation will be conducted by the project advisor via demonstrations, document reviews, and user testing sessions.

Table 4.4 outline some of the main acceptance criteria and the associated objective measure.

Criterion	Objective Measure
Functional Completeness	All features (registration, maps, booking, job posting, forum, library) operate as specified in the SRS document.
Accessibility Compliance	Platform meets WCAG 2.1 Level AA accessibility standards.
System Stability	Runs continuously without critical errors during testing and pilot usage.
Security & Data Protection	All sensitive data is encrypted and role-based access control is implemented.
Performance	Response time under normal load is 3 seconds or less for main functions.
User Satisfaction	More than 85% positive feedback from pilot users via survey or live trial.
Documentation Quality	Complete and accurate user manuals, technical documentation, and deployment guides are delivered.

Table 4.4 – Product Acceptance

5. Supporting Process Plans

5.1 Documentation

This part describes the listing of documents to be provided for the Sanad project, the applicable templates or standards for each document, and the individuals responsible for preparing and/or reviewing each document. All project documentation will adhere to recognized IEEE standards to promote consistency, clarity, and quality of deliverables.

All project documents will be prepared jointly by the development team and reviewed by Project Advisor, Dr. Enas El -Sharawy, in order to ensure they conform to the accepted academic and professional standards.

Document Name	Template	Prepared By	Review By
Submit Project Management Plan (SPMP)	IEEE1058 - SPMP	All members	Dr.Enas EI-Sharawy
Submit Project Requirements (SRS)	IEEE830 - SRS	All members	Dr.Enas EI-Sharawy
Submit Project Design (SDS)	IEEE SDS & IEEE STANDARD 1016	All members	Dr.Enas EI-Sharawy
Submit a status report 2	Use template	All members	Dr.Enas EI-Sharawy
Submit Project Test Plan (STP)	IEEE 829-1998	All members	Dr.Enas EI-Sharawy

Table 5.1 – Documentation

6. Additional Plans

In support of the functionality and safety of the Sanad system after it is deployed, the following plans for supporting processes have been established:

- **Safety, Privacy, and Security Plan:**

This states that secure authentication, data encryption, and privacy protection will be implemented in accordance with WCAG, and institutional security policies.

- **Installation and Transition Plan:**

Includes transitioning to a secure cloud environment for deployment, setting up databases, configuration management, and validating accessibility.

- **User Training Plan:**

This will create a framework for providing accessible formats for training, such as user manuals, video tutorials, and live demonstration sessions for use by users with disabilities, families and service providers.

- **Integration Plan:**

Covers system integration with external APIs and services such as Google Maps, payment gateways and assistive technology tools.

- **Maintenance and Support Plan:**

Defines periods for system updates, bug fixing, re-evaluating accessibility, and ongoing improvements based on user feedback.

Collectively, these supporting process plans will keep the Sanad platform secure, sustainable and user-friendly throughout its operational life cycle.